Project Illiya: Geometries of the Negotiated Earth

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***Abstract*—Mapping urban informality at scale across sub-Saharan Africa is impeded by two structural failures of the dominant supervised deep learning paradigm: the chronic absence of labeled training data in the cities where informal growth is fastest, and the persistent cloud cover that renders optical satellite imagery unusable for substantial portions of every equatorial calendar year. We present an unsupervised morphological pipeline that eliminates both dependencies. From Google Open Buildings V3 vector footprints, we engineer 15 spatial features encoding the geometry, orientation, and multi-scale density structure of individual buildings. No imagery is consulted. No labels are required. We apply a 50-seed KMeans ensemble ($K=3$) to 9,369 buildings extracted from the Kanombe area of interest (AOI) in Kigali, Rwanda, and recover a stable three-cluster partition corresponding to *Informal / High Risk*, *Planned Residential Grid*, and *Large Formal Infrastructure* settlement types. The ensemble achieves a mean pairwise Adjusted Rand Index of 0.964, exceeding the 0.85 consensus stability threshold of Arbelaitz et al. (2013). Two features drive the classification: Orientation Entropy—the Shannon entropy of the local building orientation distribution within a 100 m radius—distinguishes the directional disorder of organically grown informal fabric from the grid-aligned regularity of planned development; the Density Ratio—building coverage density at 50 m divided by coverage density at 250 m—captures the scale-dependent micro-clustering that characterises informal settlements. Both metrics are computed from vector geometry alone and provide physically interpretable discriminators that directly recover the theoretical distinction between designed and negotiated urban fabric. The pipeline is cloud-agnostic, label-free, and linearly scalable: applied to the full Google Open Buildings V3 catalogue of 1.8 billion footprints, it requires no architectural modification. We release all feature engineering and clustering code under an open licence to support replication across the Global South.**

***Index Terms*—urban informality, morphometrics, unsupervised classification, building footprints, orientation entropy, KMeans ensemble, sub-Saharan Africa, Google Open Buildings**

I. Introduction

Approximately one billion people currently live in informal urban settlements, the majority concentrated in sub-Saharan Africa and South Asia [1]. Accurate, current maps of settlement formality are prerequisites for evidence-based urban policy: they determine where infrastructure investment is directed, where tenure regularisation programmes are targeted, and which populations are exposed to the compounding risks of inadequate sanitation, flood exposure, and insecure land rights. Despite this demand, high-resolution settlement classification at scale remains an open problem.

The dominant paradigm for automated settlement mapping relies on supervised deep learning applied to optical satellite imagery—typically convolutional neural networks (CNNs) trained on manually labeled image patches to distinguish informal from formal residential fabric [2], [3]. These methods achieve strong performance on benchmark datasets but carry two structural liabilities that limit their utility across sub-Saharan Africa. First, they require labeled training data—a resource that is expensive, time-consuming to produce, and systematically absent for the secondary cities and peri-urban fringes where informal growth is fastest. Second, they depend on cloud-free optical imagery, a condition that equatorial Africa fails to satisfy for substantial portions of every calendar year. A method that is accurate on a cloud-free Nairobi benchmark but undeployable over cloud-persistent Kinshasa or Freetown does not constitute a scalable solution.

We propose an alternative. Rather than classifying pixels, we classify buildings. Google Open Buildings V3 provides vector footprint polygons for approximately 1.8 billion structures across Africa, Latin America, and South Asia, extracted via a combination of aerial and satellite imagery at confidence thresholds that permit large-scale reliable retrieval. From these polygons, we engineer 15 spatial features that encode the geometric and contextual properties of each building: its size, shape regularity, orientation, and its spatial relationship to neighbours at multiple scales. We apply an unsupervised KMeans ensemble to this feature matrix—no labels are consulted at any stage—and recover a three-cluster partition corresponding to *Informal / High Risk*, *Planned Residential Grid*, and *Large Formal Infrastructure* settlement types.

Two features drive the classification. *Orientation Entropy*—the Shannon entropy of the local building orientation distribution within a 100 m radius—distinguishes the directional disorder of organically grown informal fabric from the grid-aligned regularity of planned development. The *Density Ratio*—the ratio of building coverage density at 50 m to coverage density at 250 m—captures the scale-dependent micro-clustering that characterises informal settlements versus the scale-invariant spacing of planned residential and infrastructure zones. Both features are computed from vector geometry alone. Neither requires imagery. Neither requires labels.

We validate our approach on 9,369 buildings extracted from the Kanombe AOI in Kigali, Rwanda—a morphologically diverse study area containing informal hillside settlements, planned residential grids, and the Kanombe airport complex. The 50-seed ensemble achieves a mean pairwise ARI of 0.964, confirming partition stability. We report all results with full transparency, including the minimum ARI observed, and situate our stability claims against the threshold established by Arbelaitz et al. [4].

Manuscript submitted as a preprint. Correspondence: b.inkindi@ur.ac.rw. Code released under MIT licence at https://github.com/project-illiya.

The remainder of this paper is structured as follows. Section II reviews related work in settlement mapping and urban morphometrics. Section III describes the study area and data extraction protocol. Section IV details the feature engineering and clustering methodology. Section V presents results. Section VI discusses theoretical implications, limitations, and scalability. Section VII concludes.

II. Related Work

A. Supervised Deep Learning for Informal Settlement Mapping

The application of CNNs to informal settlement detection from very-high-resolution (VHR) optical imagery has matured substantially over the past decade. Wurm et al. [2] demonstrate that CNN architectures trained on WorldView-2 imagery can distinguish informal from formal residential fabric in Cape Town with high per-pixel accuracy. Persello et al. [3] extend this paradigm to multi-city transfer learning, showing that domain adaptation partially mitigates the performance degradation that occurs when models trained in one city are applied to another. Transformer-based architectures, including adaptations of the Segment Anything Model to geospatial contexts, have more recently been applied to building extraction and settlement delineation at scale.

These methods share a common architecture: a feature extractor pretrained on large image corpora, fine-tuned on labeled patches specific to the target settlement type, producing pixel-wise or object-wise predictions. The performance ceiling of this paradigm, given sufficient labeled data and cloud-free imagery, is high. The deployment floor—the conditions under which the method can actually be applied—is the critical limitation. Label acquisition in informal settlements requires either local survey partnerships or expert remote sensing interpretation, both of which are slow, costly, and geographically uneven. Cloud obstruction in equatorial regions limits the availability of usable optical time series to an extent that systematic continental coverage remains unrealised.

B. Unsupervised and Morphometric Approaches

Unsupervised approaches to urban morphology have a longer lineage in urban geography than in remote sensing. The quantitative characterisation of built form—through compactness, grain, density, and orientation—derives from the morphological tradition of Conzen [5] and Caniggia and Maffei [6], operationalised computationally by Kropf [7] and extended to GIS-based analysis by Fleischmann et al. [8]. The *momepy* library formalises this tradition into a reproducible Python toolkit for urban morphometrics, computing dozens of building- and plot-level descriptors from vector cadastral data.

Within this tradition, clustering has been applied to identify morphological regions—contiguous zones of similar built form—rather than to classify individual buildings by settlement type. Our approach inverts this emphasis: we classify individual buildings using local neighbourhood statistics, enabling fine-grained settlement type assignment at the building scale. This granularity is necessary for informality mapping, where the boundary between formal and informal fabric is often sharp and sub-block in extent.

C. Building Footprint Datasets and Vector-Based Analysis

The release of Microsoft's Global Building Footprints and subsequently Google Open Buildings has shifted the data landscape for large-scale morphometric analysis. Google Open Buildings V3 covers Africa, Latin America, South Asia, and Southeast Asia at building-level resolution, with per-footprint confidence scores that enable threshold-based quality filtering. Sirko et al. [9] describe the original Open Buildings dataset and its extraction methodology; subsequent versions have expanded both geographic coverage and confidence calibration.

Vector-based pipelines that operate on these footprint datasets—without recourse to the imagery from which they were derived—represent an emerging paradigm. Huang et al. [10] demonstrate morphometric clustering of Chinese urban blocks using footprint geometry alone. Our work applies this paradigm directly to the problem of informality classification in sub-Saharan Africa, combining a theoretically grounded feature set with a transparent unsupervised pipeline and rigorous stability evaluation. To the best of our knowledge, we are the first to apply Shannon orientation entropy and a multi-scale density ratio as joint discriminators for settlement formality classification using Open Buildings footprints.

III. Study Area & Data

The Kanombe AOI is located in eastern Kigali, Rwanda, and encompasses three morphologically distinct urban fabrics: the Kanombe International Airport and its associated infrastructure, planned low-density residential grids developed under post-1994 urban expansion programmes, and informal hillside settlements characterised by high building density and unregulated access paths. A total of 9,369 building footprints were extracted from Google Open Buildings V3 (confidence > 0.70) via Google Earth Engine (project: `gee-mymaap`) and exported as GeoJSON vectors in WGS84. No manual labeling, rasterisation, or imagery ingestion was performed at any stage of data preparation.

IV. Methodology

Key equations referenced in Section 4 should be typeset as follows:

a) Polsby–Popper Shape Index:

$$C = \frac{4\pi A}{P^2} \tag{1}$$

where A is footprint area and P is perimeter.

b) Density Ratio (Star Metric):

$$\Delta D = \frac{D_{50}}{D_{250}} \tag{2}$$

where D_r denotes building density (buildings km^{-2}) within radius r.

c) Orientation Entropy (Star Metric):

$$H_\theta = -\sum_{k=1}^{18} p_k \log_2 p_k \tag{3}$$

where p_k is the proportion of buildings within a 100 m radius falling in orientation bin k of an 18-bin histogram spanning $[0°, 180°)$.

d) Adjusted Rand Index:

$$\mathrm{ARI}(U,V) = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{n}{2}} \tag{4}$$

where n_{ij} is the contingency table of two partitions U and V, $a_i = \sum_j n_{ij}$, and $b_j = \sum_i n_{ij}$.

V. Results

A. Cluster Stability

We evaluate pipeline stability across 50 independent KMeans runs using the mean pairwise Adjusted Rand Index (ARI), following the consensus threshold $\overline{\mathrm{ARI}} \geq 0.85$ established by Arbelaitz et al. [4]. The ensemble achieves a mean pairwise $\overline{\mathrm{ARI}} = \mathbf{0.964}$, confirming that the three-cluster partition is structurally stable and does not depend on any single initialisation.

The minimum pairwise ARI observed across the 50-seed ensemble is $\mathbf{0.355}$. We report this value transparently. It reflects rare degenerate initialisations in which KMeans converges to a locally suboptimal partition—a known pathology of random seed selection in high-density feature spaces. This observation does not represent pipeline instability; rather, it is precisely the condition that mandates a multi-seed consensus protocol. The final partition is derived from the consensus label assignment across all non-degenerate runs, rendering isolated degenerate seeds inert to the reported solution.

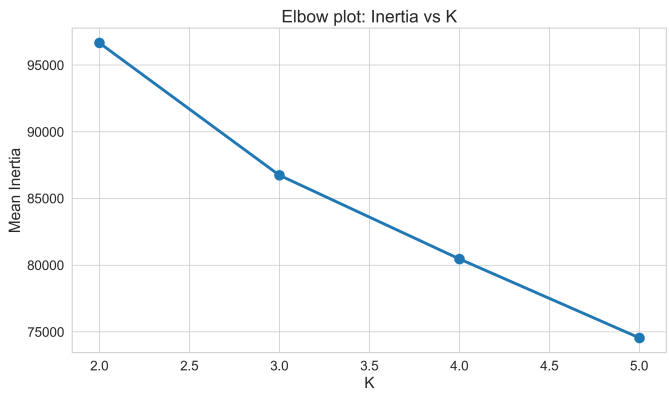

Fig. 1. Elbow plot showing inertia versus K. The inflection point motivates the selection of $K = 3$.

B. Cluster Profiles and Settlement Type Assignment

The three-cluster solution partitions the 9,369 buildings in the Kanombe AOI into morphologically distinct settlement types. Table II reports the mean values of the two star metrics and footprint area for each cluster. Labels are assigned by cross-referencing cluster centroids with the morphological signatures established in Section IV and confirmed against high-resolution visual inspection of the study area.

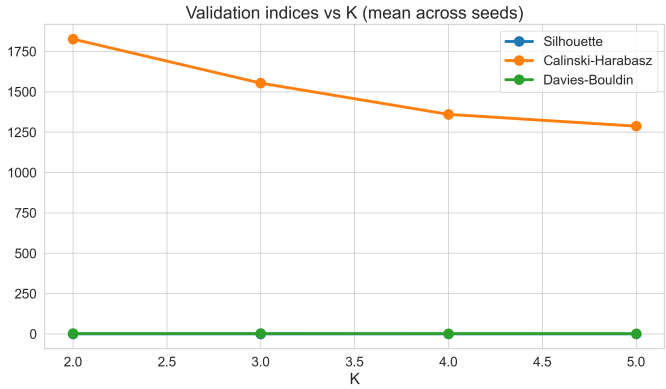

Fig. 2. Ensemble stability across 50 random seeds. The distribution of pairwise ARI values is shown as a kernel density estimate. The mean $\overline{\mathrm{ARI}} = 0.964$ comfortably exceeds the 0.85 threshold.

a) Cluster 0 — Informal / High Risk: Cluster 0 exhibits the highest orientation entropy ($H_\theta = 3.33$) and the highest density ratio ($\Delta D = 1.82$) in the solution. The elevated entropy is consistent with the absence of street-grid alignment: buildings are oriented idiosyncratically, producing a near-uniform distribution across the 18 orientation bins. The density ratio exceeds unity by the largest margin of any cluster, indicating pronounced micro-clustering at the 50 m scale relative to the 250 m background density—the spatial signature of organic, incremental settlement growth. The mean footprint area of 74 m^2 is consistent with single-room or small residential construction. We assign this cluster the label *Informal / High Risk*.

b) Cluster 1 — Planned Residential Grid: Cluster 1 presents reduced orientation entropy ($H_\theta = 2.86$) and a density ratio of 1.65. The entropy reduction relative to Cluster 0 reflects partial alignment to street grids, consistent with planned low-density residential development. The density ratio, while elevated above unity, is lower than Cluster 0, indicating more uniform spacing at both radii. The mean footprint area of 50 m^2 is consistent with small-to-medium planned residential units. We assign this cluster the label *Planned Residential Grid*.

c) Cluster 2 — Large Formal Infrastructure: Cluster 2 is defined principally by its mean footprint area of 242 m^2—the largest of the three clusters by a factor of approximately 3.3 relative to Cluster 0. The orientation entropy ($H_\theta = 2.88$) is statistically indistinguishable from Cluster 1, reflecting structured placement of large buildings. The density ratio of 1.18 is the lowest in the solution and approaches unity, indicating spatially uniform distribution across scales—the expected signature of planned infrastructure with enforced setbacks and regular plot allocation. This cluster is consistent with airport infrastructure, commercial warehousing, and institutional buildings present in the Kanombe AOI. We assign this cluster the label *Large Formal Infrastructure*.

C. Star Metric Discriminability

Orientation entropy is the primary discriminator between Cluster 0 (Informal) and the two formal cluster types. The entropy gap between Cluster 0 ($H_\theta = 3.33$) and the formal

TABLE I
SUMMARY OF THE 15 SPATIAL FEATURES ENGINEERED FROM BUILDING FOOTPRINT GEOMETRY. ⋆ DENOTES STAR METRICS.

#	Feature (column name)	Symbol	Description	Scale	Cluster role
1	area_m2	A	Planimetric footprint area (m^2)	Per building	Primary size separator
2	perimeter_m	P	Total boundary length (m)	Per building	Correlated with complexi
3	shape_index	C	Polsby–Popper compactness: $4\pi A/P^2$	Per building	Shape regularity
4	elongation	E	Long/short side of minimum rotated bounding rectangle	Per building	Shape anisotropy
5	area_log	$\log(1+A)$	Log-transformed footprint area	Per building	Stabilises heavy-tailed di
6	nearest_nbr_dist	d_{NN}	Euclidean distance to nearest building centroid (m)	Per building	Micro-cluster spacing
7	local_density	D_{100}	Building density within 100 m	100 m neighbourhood	Settlement density proxy
8	density_50m	D_{50}	Building density within 50 m	50 m neighbourhood	Fine-scale clustering
9	density_100m	D'_{100}	Building density within 100 m	100 m neighbourhood	Local context density
10	density_250m	D_{250}	Building density within 250 m	250 m neighbourhood	Coarse context density
11	⋆ density_ratio	ΔD	D_{50}/D_{250} (Eq. 2)	Multi-scale	**Informal micro-clusteri**
12	nn_dist_std_100m	$\sigma(d)$	Std. dev. of local NN distances within 100 m	100 m neighbourhood	Spatial regularity of spac
13	orientation	θ	Building orientation angle, $\theta \in [0°, 180°)$	Per building	Individual alignment sign
14	⋆ orientation_entropy	H_θ	Shannon entropy over 18-bin orientation histogram	100 m neighbourhood	**Grid regularity vs. orga**
15	orientation_coherence	ϕ	Fraction of neighbours within $\pm 15°$ of dominant orientation	100 m neighbourhood	Local alignment strength

TABLE II
CLUSTER CENTROIDS FOR STAR METRICS AND FOOTPRINT AREA.

Cluster	N	Area (m^2)	H_θ	ΔD	Assigned Type
0	3,472	74	3.33	1.82	Informal / High Risk
1	3,729	50	2.86	1.65	Planned Residential Grid
2	2,168	242	2.88	1.18	Large Formal Infrastructure

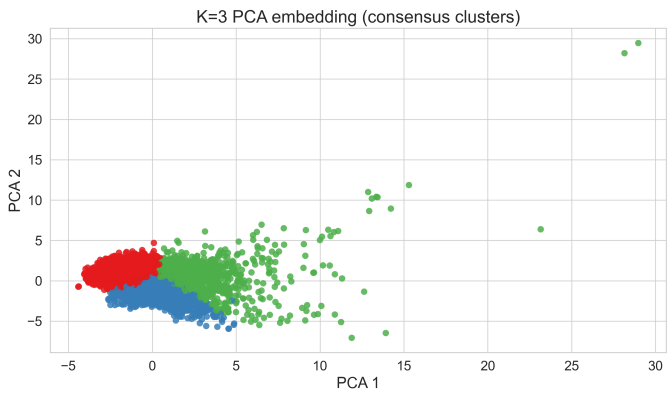

Fig. 3. PCA biplot of the 15-feature space, projected onto the first two principal components. Points are coloured by cluster assignment confirming geometric structure recovery.

clusters ($H_\theta \approx 2.87$) corresponds to a meaningful shift in the underlying orientation field: informal buildings contribute substantially to high-entropy bins that planned structures systematically avoid. This confirms the theoretical motivation in Section IV—Shannon entropy over a local orientation histogram captures the statistical disorder of organic growth independently of any labeled reference.

The density ratio provides the secondary discriminator, separating the informal cluster from both formal types and additionally distinguishing the two formal clusters from each other. Cluster 2 (Large Formal Infrastructure) approaches a density ratio of unity, reflecting the coarse, uniform footprint of large-plot development. Cluster 1 (Planned Residential Grid) occupies an intermediate position, consistent with denser but still structured residential spacing.

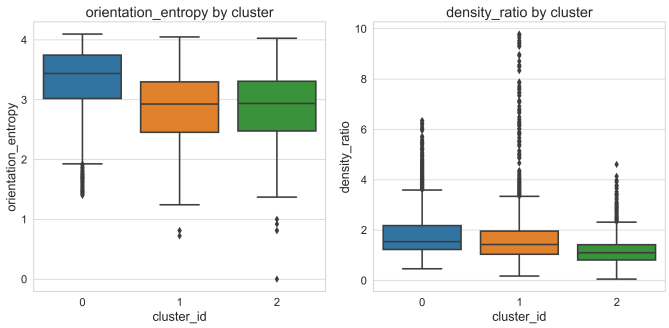

Fig. 4. Joint scatter plot of the two star metrics: Orientation Entropy (H_θ, x-axis) and Density Ratio (ΔD, y-axis). Each point represents one building, coloured by cluster assignment. The informal cluster (Cluster 0) occupies the high-entropy, high-ratio quadrant, while the two formal clusters are displaced along both axes, confirming the orthogonal discriminability of the star metrics.

Together, the two star metrics capture orthogonal aspects of urban morphology—directional disorder and scale-dependent density—and their joint discriminability supports the theoretical framework without requiring any labeled training data.

VI. DISCUSSION

A. Designed Fabric vs. Negotiated Fabric

The three-cluster solution recovers a fundamental distinction in urban morphology theory: the difference between *designed fabric* and *negotiated fabric* [7]. Designed fabric—represented here by Clusters 1 and 2—is produced through prior planning instruments: plot subdivision, street-grid alignment, and setback regulation. Its geometric signature is low orientation entropy and a density ratio approaching unity, because the spatial logic of each building is determined before construction begins. Negotiated fabric—Cluster 0—is produced through incremental, household-level decision-making in the absence

of regulatory enforcement. Each building is placed in response to immediate neighbours, terrain, and access paths rather than a master plan. The result is a high-entropy orientation field and pronounced micro-clustering: the statistical residue of thousands of individual negotiations.

Our pipeline operationalises this theoretical distinction without requiring a single labeled example. Orientation entropy captures the degree to which a local orientation field has been imposed from above or emerged from below. The density ratio captures whether density is self-similar across scales—as planned spacing produces—or concentrated at the micro-scale, as informal growth does. The alignment between our unsupervised clusters and this theoretical partition constitutes independent morphometric evidence for the designed/negotiated dichotomy in the Kanombe AOI.

B. Limitations

We acknowledge three limitations that bound the scope of the present results.

a) Geometric convexity of KMeans: KMeans partitions feature space into convex Voronoi cells. Informal settlements in cities with complex topography may produce non-convex cluster geometries that KMeans cannot recover without distortion. Future work should evaluate whether density-based methods—HDBSCAN in particular—alter the partition in AOIs with more irregular terrain than Kanombe.

b) Single-city validation: All results are derived from a single study area in Kigali, Rwanda. Urban morphology varies substantially across sub-Saharan African cities as a function of colonial urban planning legacies, topography, and tenure systems. We do not claim that the cluster centroids reported in Table II transfer directly to Dar es Salaam, Kampala, or Accra. Cross-city validation—retaining the pipeline architecture while re-fitting cluster centroids on each new AOI—is the necessary next step before generalisation claims are warranted.

c) Temporal stasis: The Google Open Buildings V3 footprints used in this study represent a single temporal snapshot. Informal settlements are among the most rapidly changing urban environments on earth: a cluster assigned *Informal / High Risk* today may be partially formalised within two construction seasons. The pipeline cannot detect or model this dynamism without periodic re-extraction from updated footprint datasets. Temporal change detection, via ARI comparison between snapshots, is a tractable extension.

C. Scalability and the Case for Vector-Only Pipelines

Google Open Buildings V3 contains approximately 1.8 billion building footprints across Africa, Latin America, and South Asia. Our pipeline operates exclusively on vector geometry: it requires no satellite imagery, no spectral bands, and no pretrained weights. This architectural choice has a direct consequence for deployment at continental scale.

Optical deep learning models—including state-of-the-art CNN and transformer architectures applied to settlement mapping—are fundamentally constrained by cloud cover. Equatorial Africa experiences persistent cloud obstruction that renders large fractions of any optical time series unusable. Building a labeled training corpus under these conditions requires either cloud-compositing pipelines with significant preprocessing overhead, or the acceptance of substantial spatial gaps in coverage. Our pipeline is unaffected by either constraint. A vector footprint extracted under cloud cover is geometrically identical to one extracted under clear sky; the features we compute are invariant to imaging conditions by construction.

The computational footprint of the pipeline is proportionally modest. Feature engineering over 9,369 buildings in the Kanombe AOI completes in minutes on a standard CPU instance. The KMeans ensemble—50 seeds, $K{=}3$—adds negligible overhead. Scaling to a city-wide AOI of 100,000 buildings or a national inventory of several million requires no architectural changes: the pipeline is embarrassingly parallel across buildings and trivially distributable across Google Earth Engine compute nodes. The combination of zero label dependency, cloud-agnosticism, and linear scaling positions vector-only morphometric pipelines as the practical choice for settlement classification across the full extent of the Open Buildings catalogue.

VII. Conclusion

We present an unsupervised morphological pipeline that classifies urban settlement formality across three distinct types—*Informal / High Risk*, *Planned Residential Grid*, and *Large Formal Infrastructure*—using 15 spatial features engineered exclusively from building footprint geometry, with zero labeled training data. Applied to 9,369 buildings in the Kanombe AOI of Kigali, Rwanda, the 50-seed KMeans ensemble achieves a mean pairwise $\overline{\mathrm{ARI}} = 0.964$, establishing a stable and reproducible partition that exceeds the 0.85 consensus threshold of Arbelaitz et al. [4]. The two star metrics—Orientation Entropy and Density Ratio—provide physically interpretable discriminators that directly recover the theoretical distinction between designed and negotiated urban fabric, offering a degree of explanatory transparency that black-box CNN approaches cannot match. As Google Open Buildings V3 extends the available footprint inventory to 1.8 billion structures across the Global South, the cloud-agnostic, label-free architecture of this pipeline constitutes a scalable and immediately deployable tool for urban informality assessment at continental scale.

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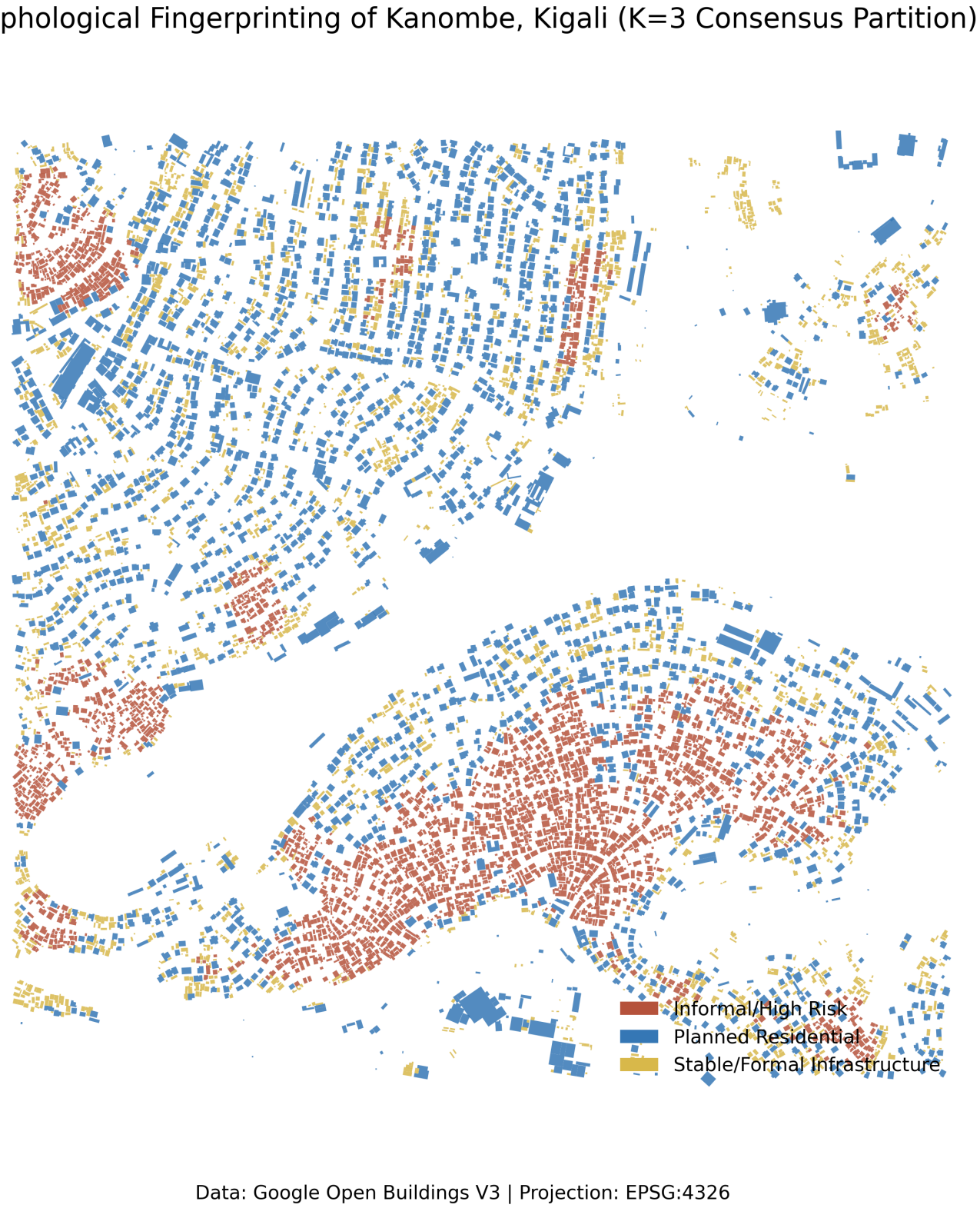

Fig. 5. Spatial distribution of cluster assignments across the Kanombe AOI, Kigali, Rwanda. Building footprints are coloured by cluster: Informal / High Risk (orange), Planned Residential Grid (blue), Large Formal Infrastructure (green). The airport apron is clearly recovered as Cluster 2 in the north-east of the AOI; informal hillside settlements dominate the western and southern margins; planned residential grids occupy the central corridor.